

Prefill Once, Ask Many: Training-Free Reversible Below-Page KV Selection for Multi-Query Long-Document Visual Question Answering

Anonymous submission

Abstract

Long-context vision-language models can answer questions about multi-page documents, but repeatedly reading the full document is costly. Existing KV-cache eviction methods address this by committing to one compressed context, even when later questions require different evidence. We present PAQ-KV (“Prefill once, Ask many Queries”), a training-free architecture for reversible, query-specific context selection in multi-query document visual question answering. PAQ-KV prefills a document once and retains its complete KV cache. For each question, it uses the frozen reader’s own attention from the question tokens to the document tokens to score 64-token page-aligned blocks, then restricts decoding to the selected blocks with a four-dimensional additive attention mask. The retained cache is neither gathered nor discarded, so every selection is reversible: each question selects anew from the same intact cache, with no additional model, retrieval index, or document re-read. Per-query re-selection outperforms committing the same scorer once by wide margins that survive a clean-cache re-run and a block-level replication. At a nominal 5% decode-attention budget ($\approx 6\%$ realized), PAQ-KV scores 0.404 on MMLongBench-Doc versus 0.328 for the trained ColQwen2 retriever, while remaining statistically indistinguishable from full-context decoding and the gold-page oracle on the development pin. A single evaluation on a fresh, disjoint, sealed 45-document set confirms the improvement ($+0.076$, with the document-clustered 95% lower bound at $+0.029$), and a development-grade third dataset, MM-DocRAG, reproduces it ($+0.057$). On LongDocURL, PAQ-KV matches but does not outperform ColQwen2 and remains slightly below full-context decoding. Porting the full method to three further readers bounds its generality: the advantage largely persists on Qwen3-VL-4B, is a statistical tie on Qwen2.5-VL-7B, and reverses on InternVL3.5-8B, tracking how well each reader’s own attention retrieves evidence below page level, while PAQ-KV stays ahead of every training-free baseline on all four readers. Retaining the complete cache while re-selecting context for each question is therefore an effective alternative to irreversible eviction for multi-query document understanding, for readers whose attention localizes evidence well.

1 Introduction

Long-context vision-language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: after reading,

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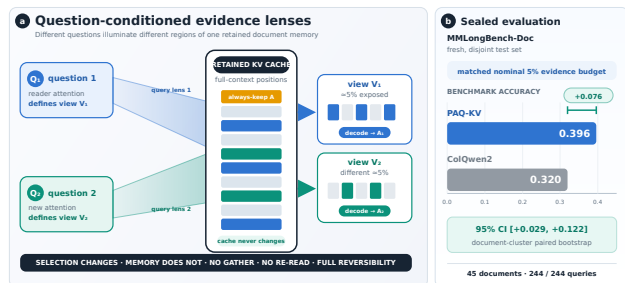


Figure 1: Method and sealed result at a glance. On fresh, disjoint MMLongBench-Doc, PAQ-KV@5% scores 0.396 versus 0.320 for corrected-loader ColQwen2 ($+0.076$, document-clustered 95% CI $[+0.029, +0.122]$; 45 documents, 244/244 queries).

the model keeps only a compact subset of its KV cache, the stored per-token state that decoding attends to, and discards the rest (Li et al. 2024; Zhang et al. 2023; Chen et al. 2024; Wan et al. 2024; Kim et al. 2025). Eviction is a single, irreversible commitment: whatever subset survives must serve every future question about that document.

In practice, however, document assistants are not asked one question. Both benchmarks in this study, MMLongBench-Doc (Ma et al. 2024) and LongDocURL (Deng et al. 2025), pose long-document visual question answering (VQA) natively as groups of questions about a shared document. Inside our pinned evaluation subsets the average document carries more than five gold-bearing questions (§4). Different questions need different evidence, so a subset committed for question 1 is stale by question 5.

This paper asks two questions. At a matched evidence budget on long multimodal documents, does *re-selecting* context for every query beat selecting once and committing? And can a training-free selector built from the reader’s own internals compete with a dedicated trained retriever? Our answer is PAQ-KV (Prefill once, Ask many Queries).

PAQ-KV works as follows. A frozen reader (Qwen3-VL-8B; Qwen Team 2025b) prefills the document once and keeps the resulting KV cache. Each question is then scored against that cache by the reader’s own attention from the question tokens to the document tokens, and those scores rank the

cache’s 64-token blocks. Blocks are the granularity below the page: a typical page spans several of them, so a block is roughly a table row, a figure caption, or a paragraph rather than a whole page of mixed content. The answer is decoded from the full cache under a four-dimensional attention mask that exposes only the top-ranked blocks. Nothing is gathered and nothing is discarded, so every position index stays the model’s own. This matters because the reader family indexes tokens with M-RoPE (Qwen Team 2025a), a multimodal rotary position embedding that gives an image token its temporal, height, and width coordinates on the page rather than a flat sequence offset; physically compacting the cache would have to reassign these composite indices, while leaving it in place avoids the question entirely. Each selection is therefore *reversible*: the next question re-selects from the same intact cache. Figure 1 summarizes the architecture and its sealed MMLongBench-Doc result.

Contributions.

- **A reversible below-page architecture and its tested mechanism.** PAQ-KV keeps the full-document prefill intact and gives every query its own below-page selection, as described above. A pre-registered gate credits the mechanism only if per-query re-selection beats two comparators: the identical scorer committed once per document, which holds the scorer fixed so the difference isolates reversibility, and a static subset chosen from the gold pages, which is not deployable but rules out the possibility that any fixed subset would do. It beats both. Re-selection adds +0.137 and +0.211 accuracy over the committed scorer (MMLongBench-Doc/LongDocURL) and clears the gold-informed static subset by +0.213 and +0.230, with document-clustered intervals excluding zero in every cell. The gate was executed with PAQ-KV itself as the fresh arm, on the full 80-document pins, post-fix code only, under a bar frozen before the run; the contrast also survives a clean-cache re-run at page level and holds when the committed scorer is a trained retriever rather than ours (§5.1).
- **An improvement confirmed out of sample, with its bounds and its failures stated.** On MMLongBench-Doc, PAQ-KV answers while attending to about 6% of the context. Given the same evidence budget, it outperforms the trained ColQwen2 retriever by 0.077, with a confidence interval that clears zero comfortably (§4.4 documents a defect we found and repaired in the ColQwen2 loader, and shows the repair does not move this comparison). The strongest evidence is a single evaluation on a sealed, disjoint set of documents, run once under a frozen harness, which reproduces the advantage almost exactly at 0.076. A development-grade third dataset, MMDocRAG, reproduces the improvement again at the same margin class (§5.4). At the same budget PAQ-KV is indistinguishable from reading the whole document, and its shortfall against an oracle handed the annotated gold pages is not statistically distinguishable from zero; the interval still permits material inferiority, so this is parity within noise rather than demonstrated equality. Two bounds run the other way. On LongDocURL, PAQ-KV

matches ColQwen2 without beating it and sits slightly below full context, which we report as a limitation rather than a second result. And two of our own pre-registered extensions failed their gates: a per-head attention mixture did not transfer, and a single sharp expert head did not improve on the diffuse block scorer (§5.6).

Evidence status. Primary full-method accuracy results use the 80-document-per-cell development pins unless explicitly labelled otherwise; the mechanism, budget, and integrity analyses use smaller screening subsets whose sizes are stated where they are used. Only the MMLongBench-Doc PAQ-KV-versus-ColQwen2 comparison was evaluated on the sealed set, once, on a fresh and disjoint 45-document cell, where the improvement holds (§5.2). The sealed LongDocURL documents remain unevaluated; the sealed run was scoped to MMLongBench-Doc. The cross-reader results of §5.5 are development-pin generality controls; no sealed claim is made for any reader other than Qwen3-VL-8B. The MMDocRAG cell, the cross-reader baseline panel, and the MMDocIR selection-quality analysis are development-grade additions with single runs and no pre-frozen bars (§5.4; Appendices B and C).

2 Related Work

Page selection and retrieval. CAPS (Zhu et al. 2026) publishes attention-based *page* selection on our benchmarks, and ColPali/ColQwen2 (Faysse et al. 2025) are the strongest trained retrieval baselines we compare against. Selecting pages with reader attention is CAPS’s method; PAQ-KV differs on the axes CAPS names as open. It selects below the page, it re-selects per query over a single retained prefill rather than committing once, and it decodes directly from the retained cache with no re-read.

KV-cache eviction. SnapKV (Li et al. 2024), H2O (Zhang et al. 2023), FastV (Chen et al. 2024), LOOK-M (Wan et al. 2024), and KVzip (Kim et al. 2025) all commit a compressed cache once. Quest (Tang et al. 2024) shows that query-aware selection can outperform query-agnostic eviction in single-query text, which motivates but does not prove our multi-query claims. We did not find prior work that combines reader-internal attention as the runtime below-page selector, end-to-end question answering at a matched budget against page-level retrieval, and long multi-query documents; we offer this as a provisional reading of the literature rather than proof of novelty.

3 Method: PAQ-KV

3.1 Setting

One long visual document, consumed as page images with no OCR, is read by a frozen VLM and will be asked k questions in sequence. The document is prefilled *once*, at the same rendering the full-context arm uses and a nominal 64K-token context target (§4 gives realized lengths), and the KV cache is retained: never evicted, never physically compressed, never gathered. The cache plays the role that a multi-vector index plays for a retriever, except that it is the reader’s own working

memory, so retrieval and reading share one representation and one model.

3.2 Per Query: Preview, Re-Select, Masked Decode

1. **Preview (score).** The tokenized question is forwarded over the retained cache, and the attention from the question rows to the context positions is aggregated into scores over 64-token, page-aligned KV blocks. Blocks at this size are fine enough to isolate a table row, a figure caption, or a paragraph, yet coarse enough for their scores to be stable. After the preview the cache is cropped back to the document boundary, so one query never contaminates the next (§4.4).
2. **Re-select.** The top-scoring blocks are kept, up to a nominal 5% decode-attention budget. The term is deliberate: the kept blocks’ keys and values were computed with full-document context during the prefill, as is standard in the Quest/SnapKV method class, so the budget bounds what decoding may attend to rather than what prefill may read. The realized attended fraction runs slightly above nominal, ≈ 5.9 to 6.0% of context on the executed screen, and is reported alongside every result. Against ColQwen2 the matched quantity is the same 5% evidence budget, counted in pages for the retriever and in context tokens for PAQ-KV, under one shared reader and answer evaluator; the two methods differ only in how they select.
3. **Masked decode.** The answer is generated while attending only to the selected blocks plus a small always-keep set: the scaffold of BOS, system, and template tokens, the question itself, and previously generated tokens. The restriction is enforced by a four-dimensional additive attention mask over batch, head, query row, and key, applied over the retained cache. There is no gathering: the K/V tensors stay at their original positions, so every M-RoPE rotary index is exactly the model’s own, and the failure mode that makes physical cache-gathering error-prone for VLMs does not arise. There is no re-read, no re-render, no second model.

Formal definition. Let Q_j be the token positions of question j , let V be the visual-content positions, and let $B_1, \dots, B_m$ be the page-aligned blocks partitioning V . For attention probability $A_{\ell h}(r, t)$ from question row r to context position t at layer ℓ and head h , the default scorer averages over all L layers, H heads, and question rows, then averages within each block:

$$\alpha_j(t) = \frac{1}{LH|Q_j|} \sum_{\ell=1}^L \sum_{h=1}^H \sum_{r \in Q_j} A_{\ell h}(r, t), \quad (1)$$

$$s_j(B_i) = \frac{1}{|B_i|} \sum_{t \in B_i} \alpha_j(t).$$

Let π_j order blocks by decreasing s_j . At budget b , selection stops at the first block whose inclusion reaches the visual-

token budget:

$$r_j = \min \left\{ r : \sum_{u=1}^r |B_{\pi_j(u)}| \geq b|V| \right\}, \quad (2)$$

$$S_j = \bigcup_{u=1}^{r_j} B_{\pi_j(u)}.$$

If $\mathcal{A}$ denotes the always-keep scaffold and G_j the generated-token positions, define $T_j(x) = \mathcal{A} \cup S_j \cup \{t \in Q_j \cup G_j : t \leq x\}$. The additive mask for decode row x is

$$M_j(x, t) = \begin{cases} 0, & t \in T_j(x), \\ -\infty, & \text{otherwise.} \end{cases} \quad (3)$$

This row-wise mask is broadcast over batch and heads. Decoding likewise ends with the cache cropped back to the document boundary, so S_{j+1} is constructed from the same retained cache.

Algorithm 1 (Appendix G) gives the loop and Figure 2 shows the mask machinery. During the preview, attention is captured by per-layer forward hooks: each layer’s attention tensor is reduced immediately and then discarded, so peak memory holds one layer’s attention at a time. Retaining all layers’ tensors, the naive implementation, runs out of memory at long contexts.

3.3 Exactness and Reversibility

A decode-identity audit, pre-registered as a kill criterion, checked the mask path against the native decode. The six audited document-query pairs span 32K to 122K tokens, a stress range deliberately wider than the evaluation pin. In all six cases an all-true mask produced zero mask-specific argmax changes relative to the native causal decode; an earlier single greedy-token divergence traced to the generation loop’s mask-independent fast-path kernel, not to the mask. We therefore claim identity in the audited cases, not in general. Because nothing is discarded, every selection is furthermore *reversible*: the next query re-scores and re-masks the same complete cache, whereas whatever commit-once eviction dropped is unrecoverable. Reversibility is the axis our evaluation isolates (§4.2): the confirmatory gate runs PAQ-KV itself against its own scorer committed once, at block level on the full pins, and the contrast clears zero in every cell (+0.137/+0.211; §5.1).

Relation to existing designs. PAQ-KV is not CAPS (a separate scoring VLM, per-page full re-forwards, page granularity, single-query); it is not coarse-to-fine page routing with a full-resolution re-read (PAQ v1, our own earlier page-level system, Appendix D); and it is not eviction (the mask is per-query and reversible). The mask route is the *accuracy* path; physically gathering the cache for savings is a separate efficiency claim, and one we do not make here.

4 Experimental Setup

4.1 Benchmarks, Pins, Reader

We evaluate on LongDocURL (Deng et al. 2025) and MMLongBench-Doc (Ma et al. 2024), both as page images,

PAQ-KV: restricted attention over an intact, position-preserving KV cache

One prefill · reader-internal scoring · per-query mask · in-place decode

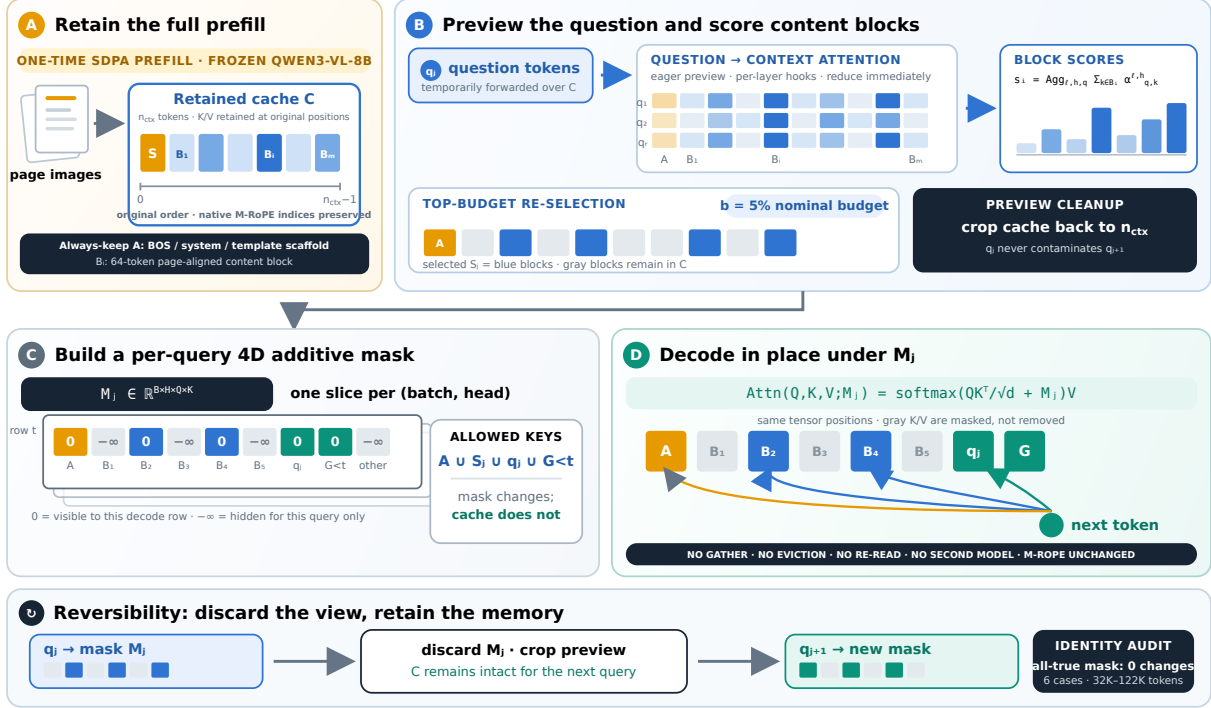


Figure 2: Detailed PAQ-KV restricted-attention mechanism. **A:** The frozen reader prefills the document once and retains the full KV cache C , including the always-keep scaffold $\mathcal{A}$ and page-aligned 64-token blocks $B_1, \dots, B_m$, at their native M-RoPE positions. **B:** For each question q_j , an eager-attention preview captures question-to-context attention using per-layer hooks. Attention mass is aggregated into block scores, and the highest-scoring blocks S_j are retained under the nominal 5% decode-attention budget. The preview is then cropped so that questions cannot contaminate subsequent queries. **C:** PAQ-KV constructs a query-specific additive mask $M_j \in \mathbb{R}^{B \times H \times Q \times K}$, assigning 0 to $\mathcal{A} \cup S_j \cup q_j \cup G_{<t}$ and $-\infty$ to all other keys. **D:** Decoding proceeds directly over the intact cache under M_j , without gathering or evicting K/V, re-reading the document, or changing positional indices. After answering, M_j is discarded, and the same complete cache is re-masked for q_{j+1} .

and add MMDocRAG (Dong et al. 2025b) as a development-grade third dataset (§5.4). For each document we build the union of the benchmark’s own-document pages across that document’s queries, dropping cross-document 64K padding, so every query’s evidence is present in one prefill. Documents qualify with at least three gold-bearing queries and at most 110 union pages, a bound that targets the nominal $\approx 64K$ -token context at read resolution (realized lengths reach $\approx 92K$ on the largest documents); selection is seed-0 frozen. The confirmatory pin holds 80 documents per cell, with 463/440 complete queries and 5.79/5.50 gold-bearing queries per document (LongDocURL/MMLongBench-Doc; Appendix G). A 30-document screen pin is its strict prefix, and results are additionally checked on the 50 new documents only (§5.1). The reader is Qwen3-VL-8B (Qwen Team 2025b), frozen throughout: scaled dot-product attention (SDPA) serves prefill and reading, and eager attention is enabled only during the per-query preview so that attention from the question tokens to the content can be captured. Page-level arms operate at a matched page budget $b \in \{2.5\%, 5\%, 10\%\}$, with the pre-registered gate budget at $b =$

5%; accuracy is the benchmark’s document-VQA scorer. The primary results are a Qwen3-VL-8B case study; §5.5 ports the full method to three further readers (Qwen3-VL-4B, Qwen2.5-VL-7B, InternVL3.5-8B) as a development-pin generality control.

4.2 Statistical Protocol

All inference uses a document-cluster paired bootstrap: queries of one document share the document and (for stale arms) the single committed subset, so bootstrapping over queries would be anti-conservative pseudoreplication. Intervals resample documents with replacement within each cell (10,000 draws, seed 0). Paired contrasts are reported as the mean difference Δ with its two-sided 95% confidence interval (CI), written $\Delta [l, u]$; bare bracketed pairs after an effect denote this interval throughout. Every pre-registered gate in this study is a criterion on the interval’s lower bound: a gate passes when the lower bound exceeds zero, or the stated margin, and where that bound alone carries the argument we name it in words. Per-cell results are primary; pooled contrasts use equal cell weighting and are labelled. Every gate

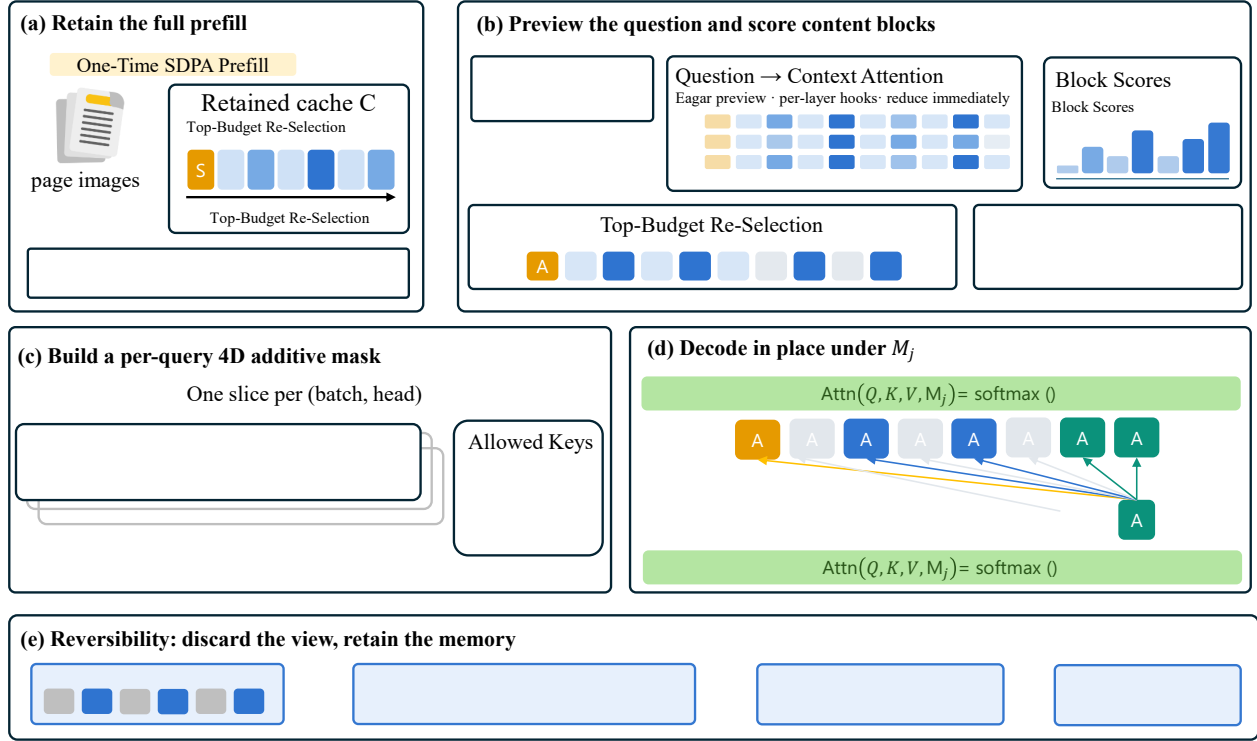


Figure 3: PAQ-KV in five steps. **(a)** The document’s page images are prefilled once with SDPA and the full KV cache C is retained, scaffold $\mathcal{A}$ included. **(b)** Each question is previewed with eager attention and per-layer hooks; attention from the question tokens to the context is reduced to scores over 64-token page-aligned blocks, and the top blocks are re-selected under the budget. **(c)** The selection becomes a per-query four-dimensional additive mask, one slice per batch and head, whose allowed keys are the scaffold, the selected blocks, the question, and the generated tokens. **(d)** The answer is decoded in place under M_j over the intact cache. **(e)** After answering, the mask is discarded and the complete cache remains, so the next question re-selects from the same memory: selection is reversible.

was frozen before its run.

The isolation gate. That query-aware selection outperforms query-agnostic eviction is both a known text result (Tang et al. 2024) and a potential strawman. We therefore fix the tested axis to *reversibility*: every baseline runs its canonical scorer but is deployed stale, committed once per document (query-aware scorers commit a min-max-normalized aggregate over the document’s first $M = \min(3, k)$ queries), while the fresh arm re-selects per query. Reversibility is credited only if the fresh arm outperforms two comparators. The first is its *own* scorer committed once: scorer and budget are identical, so the delta isolates the mechanism. The second is a strong *gold-informed* static subset, the top pages by gold-page frequency across the document’s queries: not deployable, and not provably optimal, but immune to the weak-static artifact that invalidated an earlier one-comparator gate. The gate is executed with PAQ-KV itself as the fresh arm at its native below-page granularity (the committed aggregate and the gold-frequency scores are applied over the same 64-token blocks), on the full 80-document pins of both cells, post-fix code only; the bar, arms, budget, seed, imputation policy, and sha-256 hashes of every code component were frozen and committed before the run.

4.3 Baselines

The executed panel (Appendix A) compares, per query at matched budget on the same pinned data and reader: ColQwen2 (Faysse et al. 2025) fresh and stale, CLIP (Radford et al. 2021) fresh, SnapKV fresh and stale, the isolation arms above, random and truncation controls, and a query-blind eviction family (H2O, a context-only FastV proxy, LOOK-M, and a KVzip recv_max proxy). The eviction arms are disclosed proxies that sit at or below random page recall on these cells, so the trained retriever is the binding comparison; we rest no claim on outperforming proxy eviction. Full-context and gold-page-oracle rows are references, not competitors. The training-free arm family is additionally executed on the MMDocRAG cell (Table 1) and on all three second readers (Appendix B).

4.4 Methods Integrity

An independent code review found a real bug in the per-query scoring loop used across this study: the runtime mutates the retained cache in place, and the loop reused the document prefill across a document’s queries without restoring it, so query q_i scored while attending to $q_0 \dots q_{i-1}$. The fix, cropping the cache after every per-query preview, is now applied everywhere, and its impact was quantified rather than assumed. On a 24-document, 150-query clean-versus-polluted re-run, the key isolation contrast is +0.264 clean versus +0.284 polluted, since the contrast largely cancels the bug when both arms share the same scorer. Every gate and headline number in this paper postdates the fix, including the confirmatory isolation gate of §5.1, which was re-executed in full on post-fix code with PAQ-KV as the fresh arm and verified at runtime (cache-length assertions after every preview and decode, plus an order-invariance probe per cell that

recomputes a query’s preview after all other queries and requires bit-identity). The one surviving pre-fix artifact is the historical page-level panel of Appendix A, whose absolute fresh-arm accuracy carries ≈ 0.04 inflation; it is labelled as historical and no claim rests on its absolute levels.

Baseline-loader integrity. A second independent review caught a defect in the *comparator*, not in our method: the cached ColQwen2 baseline had run with its trained final retrieval-projection LoRA (custom_text_proj) silently zeroed. `vidore/colqwen2-v1.0` is a PEFT adapter, and under our `peft/transformers` versions a key remap inserted language_model into the LoRA target path. The remap is correct for the language-model layers but wrong for the top-level custom_text_proj, so the checkpoint’s projection-LoRA tensors landed at a non-matching path and the final projection ran with base weights only. Because the other adapter layers still loaded, the baseline stayed functional and well above random, which masked the defect until we inspected the load report and the projection weights directly. We patched the loader, added a post-load assertion that the two custom_text_proj tensors match the checkpoint, and re-scored ColQwen2. The correction is immaterial to the main result. Fixed-versus-broken gold-page recall at the 5% budget is 0.539 versus 0.534. The top-5% page selections overlap at Jaccard 0.947, with only 34 of 444 queries changing any selected page. Re-reading those changed queries moves fresh-read accuracy by -0.0002 (0.3282 corrected versus 0.3284 broken), and the improvement over ColQwen2 is $+0.077$ against both loaders (95% CI lower bounds $+0.040$ corrected and $+0.039$ broken), so the main result is not loader-inflated. The corrected loader is used for the main MMLongBench-Doc comparison (development and sealed), the MMDocRAG cell, and the systems measurement; the LongDocURL result and the historical page-level panel retain the earlier loader and are labelled as such. We do not infer a direction for the correction on the cells that were not re-run: on MMLongBench-Doc the change was negligible in both directions, and LongDocURL, already a matched result rather than an improvement, is reported as an uncorrected-loader development result.

Separately, this study adopted a sealed-test discipline: the 80-document pin is development-only, and a fresh, disjoint 45-document / 244-query MMLongBench-Doc set (zero overlap with the pin) was reserved and evaluated *exactly once* under a frozen harness. The confirmatory bar, comparator, budget, imputation policy, and all eight component code hashes were frozen (and a byte-hash of the sealed pin recorded) before the single run, and a guard enforces this. That one sealed evaluation confirms the improvement out-of-sample (§5.2); the sealed LongDocURL documents were not evaluated (out of scope for this MMLongBench-scoped result).

5 Results

5.1 Isolating Reversibility

The isolation gate runs at $b = 5\%$ on all 907 pinned queries of the confirmatory pin (80 documents per cell; zero

skips), pitting PAQ-KV itself — per-query below-page re-selection over the retained cache — against its two comparators built from the same 64-token block scores. PAQ-KV scores 0.531/0.405 (LongDocURL/MMLongBench-Doc), the identical scorer committed once scores 0.320/0.268, and the gold-informed static scores 0.301/0.192. Both gate contrasts pass in both cells: fresh over commit-once by $+0.211/+0.137$, and fresh over the static by $+0.230/+0.213$ (full table in Appendix G; Figure 5 shows every executed isolation interval). The committed and fresh selections overlap at mean Jaccard 0.368, so the committed subset is not a strawman: it shares a third of the fresh arm’s blocks and still loses. All three arms realize a matched $\approx 6.0\%$ of context. Three robustness checks follow. First, an independent extension: because the screen pin is embedded in the confirmatory pin, we re-ran the statistics on the 50 new documents per cell only ($n = 586$ queries), where fresh over commit-once is $+0.180 [+0.124, +0.237]$ and fresh over the static $+0.206 [+0.161, +0.249]$ (pooled, equal-cell). Second, granularity and history: the same two-comparator gate passed at page level on this pin (its pre-fix historical run: $+0.257/+0.163$ over commit-once), and the page-level contrast survives the clean-cache re-run of §4.4 ($+0.264$, 95% CI lower bound $+0.187$), so the effect is not specific to block granularity or to the fix. Third, scorer generality: the same fresh-over-stale collapse holds for a retrieval scorer, with ColQwen2 fresh at 0.423 versus stale at 0.178 (historical page-level cohort; query-micro; equal-cell pooling agrees), so reversibility is a property of the multi-query regime rather than of our scorer. A pre-registered “commitment-pressure” mechanism gradient failed its formal interaction test (both intervals include zero) and is retained as exploratory only.

5.2 MMLongBench-Doc Results

Main comparison. On the 80-document development pin (444 queries decoded; contrasts on the 440 with cached references), PAQ-KV@5% scores 0.404 versus 0.328 for ColQwen2-fresh. The improvement of $+0.077$ (95% CI $[+0.040, +0.115]$) meets the pre-registered per-cell bar with margin (Table 2). ColQwen2 here runs with its trained retrieval-projection LoRA correctly applied; §4.4 documents the loader defect we found and corrected and shows the correction is immaterial to this comparison. Two development-pin re-slices sharpen where the improvement comes from. At the same nominal budget the retriever delivers *more* gold evidence to the reader, not less: ColQwen2’s kept pages cover 0.539 of gold-page tokens versus 0.344 for PAQ-KV’s selected blocks, yet PAQ-KV answers more accurately, so gold-evidence recall and answer accuracy dissociate. And the paired improvement concentrates on cross-page evidence ($+0.109 [+0.052, +0.170]$, versus $+0.054 [+0.001, +0.108]$ on single-page evidence), which directly supports the dispersed-evidence reading of this cell. Appendix H gives both recall notions and the full evidence-type stratification.

Out-of-sample confirmation. The improvement holds on a fresh, disjoint, sealed set evaluated exactly once. On 45 MMLongBench-Doc documents / 244 queries with zero

overlap with the development pin, under a frozen harness whose bar, comparator, budget, imputation policy, and code hashes were all pre-registered (§4.4), PAQ-KV@5% scores 0.396 vs. ColQwen2-fresh 0.320: $+0.076$ (document-clustered 95% CI $[+0.029, +0.122]$), complete (244/244, zero skips) and unchanged under worst-case missingness imputation. The pre-registered verdict (complete-case CI lower bound > 0 , with an adversarial-imputation fallback that binds only if any query is missing) is met. The sealed effect ($+0.076$) closely matches the development-pin estimate ($+0.077$); the sealed interval is appropriately wider (lower bound $+0.029$ versus $+0.040$) at the smaller held-out sample. Only the PAQ-KV-vs-ColQwen2 comparison was sealed-evaluated; the full-context and gold-page-oracle reference arms above are development-pin only and were not run on the sealed set. This single out-of-sample evaluation is the strongest evidence in the paper and directly addresses the reuse of the development pin across this study (disclosed in Appendix F). PAQ-KV attends to a nominal 5% (realized ≈ 5.9 to 6.0%) of the context; at that budget it is indistinguishable from full context ($+0.009 [-0.022, +0.040]$) and not statistically distinguishable from the gold-page oracle ($-0.026 [-0.059, +0.008]$). The latter interval permits material inferiority, so this is parity within noise rather than demonstrated equality.

Mechanism checks. On the 30-document screen (distinct from the 80-document cohort above), the result survives every mechanistic adversarial check: it is query-dependent (PAQ-KV – random-block selection = $+0.329 [+0.238, +0.418]$), query-specific (PAQ-KV – wrong-question selection = $+0.698 [+0.603, +0.784]$, where the collapse supports query-specificity), reversible (PAQ-KV – commit-once = $+0.379 [+0.273, +0.480]$; this screen control commits the *first query’s* selection, a weaker committed arm than the confirmatory gate’s first-three-query aggregate, which is why its margin exceeds the confirmatory $+0.137$), and below-page granularity helps over page level ($+0.054 [+0.011, +0.102]$). These are development-pin screen analyses with disclosed caveats: the screen budget-curve monotonicity gate failed on this cell (0.398 at 10% vs. 0.402 at 5% on the 157-query screen subset), though the subsequent full-pin sweep is monotone and attributes that failure to small-sample noise (Appendix I), and the effect they dissect is separately confirmed out-of-sample on the sealed set above.

Additional matched-budget baselines. On the same 80-document pin and reader, a BM25 page retriever scores 0.211 at the matched 5% budget, which PAQ-KV exceeds by $+0.193$ (95% CI lower bound $+0.152$; 444 queries). A training-free *page-level* CAPS-style expert-head control scores 0.356, which below-page PAQ-KV exceeds by $+0.048$ (95% CI lower bound $+0.009$; unchanged under worst-case missingness imputation). The margin over this control is real but thin, so the improvement over the trained retriever remains the stronger claim; the full-pin budget sweep, including matched-budget comparators at 2.5 and 10%, is in Appendix I. The query-blind KV-eviction family (SnapKV-stale, H2O, FastV, LOOK-M, KVzip proxies) remains in Appendix A, where it sits at or below random page selection

	LongDocURL 80 docs / 463 q	MMLongBench-Doc 80 docs / 444 q	MMLB (sealed) 45 docs / 244 q	MMDocRAG 62 docs / 585 q
PAQ-KV@5% (ours)	0.5315	0.4042*	0.396*	0.4089*
ColQwen2-fresh (matched)	0.5124	0.3282	0.320	0.3517
CAPS-style expert head (page)	–	0.3561	–	–
BM25 (page retriever)	–	0.2109	–	–
SnapKV-fresh (page) [†]	0.298	0.157	–	0.1557
CLIP-fresh [†]	0.155	0.164	–	0.1119
ColQwen2-stale (commit-once) [†]	0.216	0.138	–	–
random [†]	0.068	0.022	–	0.0368
truncation [†]	0.033	0.041	–	0.0327
full-context (ref.)	0.5456	0.3965	–	0.4018
gold-page oracle (ref.; not strict)	0.6092	0.4314	–	0.4028

Table 1: Main results: accuracy at the nominal 5% budget against the trained retriever, the training-free baselines, and the reference arms, per benchmark (reader Qwen3-VL-8B). Bold marks the best non-reference arm per column; * marks columns where the paired PAQ-KV-versus-ColQwen2 difference is statistically significant, that is, its document-clustered 95% CI excludes zero (Table 2). The LongDocURL and MMLongBench-Doc columns are the 80-document development pins; MMLB (sealed) is the fresh, disjoint 45-document set evaluated exactly once under the frozen harness, where only the PAQ-KV-versus-ColQwen2 comparison was run; MMDocRAG is the development-grade 62-document fresh-documents pin of \$5.4 (single run, no pre-registered bar). [†]On LongDocURL/MMLongBench-Doc these rows come from the page-level executed panel (903-query cohort, Appendix A); they read selected pages rather than masked KV and predate the cache fix of \$4.4, so they are context rather than same-cohort comparisons. The MMDocRAG column’s baselines share PAQ-KV’s frozen stack. MMLongBench-Doc decodes 444 queries; paired contrasts use the 440 with cached references (the 4 mi_phone rows lack cached references). Realized budgets on MMLongBench-Doc: PAQ-KV attends a mean 6.0% of the prefill context (median 6.0, range 5.5 to 7.5%), including the always-attended non-visual prefix; ColQwen2 keeps whole pages, a mean 4.6% of pages under whole-page rounding.

Δ (95% CI)	LongDocURL	MMLongBench-Doc	MMLB (sealed)	MMDocRAG
PAQ-KV – ColQwen2-fresh	+0.019 [−0.019, +0.059]	+0.077 [+0.040, +0.115]	+0.076 [+0.029, +0.122]	+0.057 [+0.040, +0.076]
PAQ-KV – full-context	−0.014 [−0.031, +0.003]	+0.009 [−0.022, +0.040]	–	+0.007 [−0.004, +0.018]
PAQ-KV – gold-page oracle	−0.078 [−0.116, −0.040]	−0.026 [−0.059, +0.008]	–	+0.006 [−0.009, +0.022]
Pre-registered bar (lower bound > 0)	NOT MET	MET	MET	no bar frozen

Table 2: Key paired contrasts for Table 1: mean paired difference Δ with its document-clustered bootstrap 95% confidence interval (10,000 draws; \$4.2). The pre-registered per-cell bar passes when the CI lower bound exceeds zero. On LongDocURL PAQ-KV is below full context, with an interval that includes zero, and below the gold-page oracle, with an interval that excludes zero; on MMLongBench-Doc and MMDocRAG it is statistically indistinguishable from both references. Only the PAQ-KV-versus-ColQwen2 contrast exists on the sealed set; the MMDocRAG cell is development-grade, so no bar was frozen there.

(\$4).

5.3 LongDocURL Results

On the 80-document development pin (463 queries), PAQ-KV@5% posts a positive point estimate over ColQwen2-fresh of +0.019 (95% CI [−0.019, +0.059]), which does not meet the pre-registered bar (Table 2). On this cell PAQ-KV also sits below full context, with an interval that includes zero, and clearly below the gold-page oracle. The mechanism checks, query dependence and reversibility, pass here too; what is missing is the margin. We treat this as a bounding observation on where below-page selection helps rather than a second main result, and defer the conversion-bound decomposition to Appendix J.

5.4 MMDocRAG: A Third Dataset

To test the dispersed-evidence reading beyond the two primary benchmarks, we added MMDocRAG (Dong et al. 2025b), a retrieval-augmented document-QA dataset with cross-modal, multi-page evidence chains (52% of queries carry cross-page evidence). MMDocRAG is corpus-adjacent to our pins, so a document-name audit ran first: every document overlapping our development or sealed pins was excluded before any experiment, leaving a fresh 62-document, 585-query pin. Appendix F details the audit and this pin’s protocol deviations; the cell is development-grade, with a single run, no pre-frozen bar, and a document count below the study’s 80-document floor.

The improvement over the trained retriever reproduces. PAQ-KV@5% scores 0.409 versus 0.352 for corrected-loader ColQwen2 at matched budget (Table 1). The margin of +0.057 (95% CI [+0.040, +0.076]; Table 2) is the same

margin class as the MMLongBench-Doc development and sealed results. PAQ-KV is again statistically indistinguishable from its own full-context read and from the gold-page oracle, and the oracle here barely exceeds full context, which is what genuinely dispersed evidence predicts: handing the reader only the annotated gold pages does not help it. The training-free baselines fall far behind (SnapKV-fresh 0.156, CLIP-fresh 0.112; Table 1), mirroring the panel ordering of Appendix A. The dispersed-versus-conversion-bound hypothesis of the conclusion therefore survives its first test on a dataset outside the original pair, at development grade.

5.5 Cross-Reader Generality

All results above use one reader. To test whether the effect is specific to Qwen3-VL-8B, we ported the complete method (reader-own attention scoring, block selection, masked decode) to three further VLMs: Qwen3-VL-4B (the same architecture at half the parameters), Qwen2.5-VL-7B (Qwen Team 2025a) (the prior generation of the same family), and InternVL3.5-8B (Wang et al. 2025) (a different VLM whose language backbone is Qwen3). Every port had to pass the same all-true-mask decode-identity gate (teacher-forced argmax parity against the native causal decode, plus a restrictive mask that changes the output, on documents spanning at least 30 pages) before any accuracy number was read. Table 3 reports both development cells.

The outcome is a gradient ordered by architectural distance from the development reader. Qwen3-VL-4B largely preserves the effect (+0.030 [−0.004, +0.065] on MMLongBench-Doc) and, like the 8B, scores above its own full context (0.3640 vs. 0.3480). Qwen2.5-VL-7B is a statistical tie (+0.007 [−0.037, +0.049]). InternVL3.5-8B falls significantly below its comparator (−0.067 [−0.106, −0.029]) and below its own full context (−0.076 [−0.113, −0.041]). On LongDocURL every reader sits at or below its MMLongBench-Doc contrast with the ordering preserved, consistent with that cell being conversion-bound (§5.3); against its own full context there, Qwen3-VL-4B is indistinguishable (−0.006 [−0.027, +0.017]) while Qwen2.5-VL-7B and InternVL3.5-8B fall short (−0.060 [−0.100, −0.021] and −0.077 [−0.109, −0.047]). A pre-registered development-only tuning study on InternVL (budgets {5, 10, 20, 30}%, four attention read-outs, tuned on 40 documents and evaluated on the held-out 40) found budget to be the only effective lever: the deficit narrows monotonically to −0.001 at a 30% budget, but the lower bound never clears zero at any budget, and the tuned read-out adds nothing beyond budget (+0.016, 95% CI lower bound −0.014, paired against the default), so the negative is not an artifact of the 5% budget or of the default attention read-out.

The gradient tracks a measurable property: how well the host reader’s own attention retrieves evidence below page level. On the queries a reader answers correctly from its own full context, its 5% selection recovers 0.88 (Qwen3-VL-8B), 0.87 (Qwen3-VL-4B), 0.70 (Qwen2.5-VL-7B), and 0.53 (InternVL3.5-8B) of that accuracy. Since the effect survives halving the reader within the same architecture but reverses across families, we read the boundary as architectural rather than a matter of scale, and state the scope condition plainly:

PAQ-KV is a white-box method whose benefit is contingent on the host reader’s attention being an effective below-page retriever. Three caveats attach. These are development-pin generality controls, not sealed claims. The two non-Qwen3-VL readers run at reduced realized context because of their context ceilings, so only within-reader contrasts are interpreted. And InternVL3.5-8B’s language backbone is Qwen3, which is why the mask machinery transfers at all; it is a cross-VLM test but not a fully independent language architecture. Three further candidates (Molmo-7B-D, MiniCPM-V-4.5, PaliGemma-2) could not run the method under the pinned software stack (remote-code version drift for the first two; an ≈8K context that cannot hold a multi-page document for the third) and are reported as incompatible rather than forced, which maps the method’s practical deployment envelope: a version-current implementation exposing hookable attention and an honored 4D additive mask, and enough context to pre-fill a whole document.

Baseline family on the second readers. The training-free arm family of Appendix A was additionally executed on all three second readers at the same budget and page-set semantics (Appendix B). Three regularities stand out. First, PAQ-KV stays ahead of every training-free baseline on every reader, including the readers where it loses to the trained retriever: SnapKV-fresh, the strongest such baseline, sits significantly below PAQ-KV in all six reader-cell combinations. Second, SnapKV-fresh here is exactly the reader’s own question-preview attention selecting pages, so its absolute level reads out each reader’s attention-as-retriever quality; it falls across readers in the same order as the recovery gradient above, so the selection signal itself degrades in that order, independently of the block machinery. Third, committing either scorer once collapses it on every reader, so the multi-query staleness penalty is a property of the regime rather than of the development reader. On InternVL3.5-8B the gold-page oracle additionally exceeds the reader’s own full-context accuracy, so that reader’s bottleneck is finding evidence rather than converting it: where the host’s attention retrieves poorly, a stronger selector recovers real accuracy that PAQ-KV leaves on the table.

5.6 Negative Results

Per-head mixture transfer gate. No LoRA adapter was ever trained: this pre-frozen check failed first and stopped that line of work. We fit a logistic per-head mixture over all $36 \times 32 = 1,152$ (layer, head) question-to-page attention channels on off-benchmark provenance-checked data. The transfer gate fails: mixture gold-page recall is 0.652/0.594 vs. 0.676/0.616 for the single head, with pooled mixture − single = −0.023 [−0.037, −0.008] (907 queries), so the mixture overfits and this direction was closed. The best zero-benchmark-contact head (L21.H18) still coincides with the benchmark-supervised expert layer.

Sharp expert head as block scorer. Replacing PAQ-KV’s diffuse mean-attention block scorer (V0) with the sharp head L21.H18 (V1) fails the pre-frozen gate: on LongDocURL it only nudges the point estimate (0.5315 → 0.536; V1 −

ColQwen2 = +0.024 [−0.012, +0.066]) and does not separate from V0 (the V1–V0 95% CI lower bound is −0.011, n.s.), while on MMLongBench-Doc it hurts (0.402 → 0.381 on the screen documents). As elsewhere in this study, recall-optimized selection is not accuracy-optimized selection.

5.7 Systems Measurements

We measured the deployment profile of the true PAQ-KV path (below-page masked decode over the *retained* full-document KV cache) against ColQwen2-fresh and full context on an 8-document, 31-query systems subset of the pin (single GPU, frozen configuration; Table 4). This corrects an earlier internal accounting that timed a different page-routing variant. The honest summary runs against us in two of three respects. Per online query PAQ-KV is roughly $2.25\times$ slower than the retriever pipeline. It is also far heavier in resident state, since it retains a multi-gigabyte per-document cache where the retriever stores a megabyte-scale page index. Against re-reading the document for every question it is roughly $11\times$ faster, and the one-time prefill amortizes after about one query per document. The efficiency claim is therefore simplicity and deployment shape, not speed: no separately trained retriever checkpoint, no stored vector index, one deployed model, and reversible query-time selection. Table 4 gives the component costs and the amortized accounting.

6 Conclusion

PAQ-KV turns a frozen VLM’s own KV cache into a queryable, reversible evidence store: prefill once, then per query re-select below-page blocks and decode under a restricted-attention 4D mask over the intact cache, with no gathering and M-RoPE preserved. Reversibility is the property this design provides, and it holds across scorers, granularities, and readers. On dispersed-evidence long-document VQA the full method outperforms a state-of-the-art trained visual retriever at matched budget and is statistically indistinguishable from the gold-page oracle at $\sim 6\%$ of context, with the mechanism adversarially audited. The improvement is confirmed out-of-sample on a sealed, disjoint held-out set evaluated exactly once (+0.076 with a positive document-clustered lower bound), and it reproduces at development grade on a third dataset, MMDocRAG (§5.4). Where the method does not outperform the retriever (LongDocURL) we report the matched result as a limitation and characterize the cell as apparently conversion-bound. A cross-reader port to three further VLMs bounds generality: the advantage survives halving the reader within the same architecture, narrows to a tie on the prior Qwen-VL generation, and reverses on InternVL3.5-8B, so the benefit is contingent on the host reader’s own attention being an effective below-page retriever. We offer the dispersed-versus-conversion-bound split as a hypothesis that survived its first test outside the original benchmark pair and now needs a sealed-grade confirmation.

Method	LDU	MMLB
Qwen3-VL-8B (development reader; ~ 69 K ctx)		
PAQ-KV@5% (ours)	0.5315	0.4042
ColQwen2 (matched)	0.5124	0.3282
full-context (ref.)	0.5456	0.3965
Δ (95% CI)	+0.019 [−0.019, +0.059]	+0.077 [+0.040, +0.115]
Qwen3-VL-4B (same architecture; ~ 69 K ctx)		
PAQ-KV@5% (ours)	0.4966	0.3640
ColQwen2 (matched)	0.4908	0.3336
full-context (ref.)	0.5024	0.3480
Δ (95% CI)	+0.006 [−0.028, +0.043]	+0.030 [−0.004, +0.065]
Qwen2.5-VL-7B (same family, prior gen.; ~ 52 K ctx)		
PAQ-KV@5% (ours)	0.3953	0.2698
ColQwen2 (matched)	0.4728	0.2623
full-context (ref.)	0.4558	0.3237
Δ (95% CI)	−0.078 [−0.116, −0.037]	+0.007 [−0.037, +0.049]
InternVL3.5-8B (cross-family, Qwen3 LM; ~ 36 K ctx)		
PAQ-KV@5% (ours)	0.2186	0.1876
ColQwen2 (matched)	0.3081	0.2542
full-context (ref.)	0.2961	0.2637
Δ (95% CI)	−0.090 [−0.127, −0.057]	−0.067 [−0.106, −0.029]

Table 3: Cross-reader generality at the 80-document development pins (LDU = LongDocURL, 463 queries; MMLB = MMLongBench-Doc, 444 queries). Within each reader block, PAQ-KV (the reader’s own attention selects $\approx 5\%$ of blocks) is compared with the same reader reading the top-5% pages ranked by the cached ColQwen2 scores, so every contrast is within-reader at matched budget; the reader’s own full context is a reference, not a competitor. Bold marks the better of the two matched-budget arms per column by point estimate; significance is carried by the contrast rows, which report the PAQ-KV-minus-ColQwen2 difference with its document-clustered 95% CI. Block headers give the mean realized MMLongBench-Doc prefill length: the two non-Qwen3-VL readers run below the ≈ 69 K budget because of their context ceilings, so only within-reader comparisons are interpreted. The Qwen3-VL-8B block repeats Table 1; its MMLB improvement is additionally confirmed on the sealed set (§5.2). Qwen3-VL-4B reuses byte-identical input geometry (verified on all 80 documents), so that block is a pure size ablation.

	extra params	per-doc state	cold st. /doc (s)	online /q (s)	amort. /q (s)	VRAM (GiB)
PAQ-KV (ours)	0	8.15 GiB	11.3	1.04	3.07	44.5
ColQwen2-fresh	2.246 B	11 MiB	2.5	0.46	0.91	21.8
full-context	0	n/a	n/a	11.59	11.59	44.6

Table 4: Deployment profile on an 8-document / 31-query systems subset (frozen configuration, single GPU). “Extra params” = additional retriever/selector parameters beyond the shared reader. “Cold start” is the one-time per-document cost: full-document prefill for PAQ-KV, page-embedding index build for ColQwen2. “Amort. lat./q” adds the cold start amortized at the development pin’s observed 5.55 queries/document to the online latency (at the systems subset’s own 3.9 queries/document the figures are 3.95/1.10/11.59 s). PAQ-KV’s online latency decomposes into 0.19 s selection and 0.85 s masked decode; it is $\approx 2.25\times$ slower and far heavier in resident state than ColQwen2, and $\approx 11\times$ faster than full context (cold start amortized after $N\approx 1.07$ queries/document). PAQ-KV’s online latency excludes ≈ 1.99 s/query of removable research-harness input reconstruction (component sum ≈ 3.03 s/query in the current implementation); mean generated lengths differ across arms (5.3/3.5/4.1 tokens for PAQ-KV/ColQwen2/full-context). The advantage is 0 extra retriever parameters, no index, one model, and reversible query-time selection, not speed.

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A Full Executed Panel

Table 5 reports every executed arm at the gate budget on the confirmatory pin. This is the historical page-level panel: its fresh arm predates the cache fix of §4.4 (≈ 0.04 absolute inflation) and its retrieval rows use the uncorrected ColQwen2 loader; it is retained as context for the arm family’s ordering, and no headline claim rests on its absolute levels. The query-blind eviction family sits at or below random gold-page recall on these cells and is included for completeness (§4). Secondary pooled statistics: the fresh page-level selector exceeds the best arm of that family (SnapKV-stale) with min-over-family 95% CI lower bound $+0.202$; fresh – SnapKV-fresh = $+0.094$ [$+0.069, +0.122$]; fresh – oracle = -0.199 [$-0.239, -0.161$]. Gold-page recall at $b=5\%$ (LongDocURL / MMLongBench-Doc): fresh page selector 0.494/0.443; gold-informed static 0.381/0.373; same-scorer commit-once 0.137/0.148; ColQwen2-fresh 0.664/0.536.

B Cross-Reader Baseline Panel

The cross-reader table of §5.5 compares PAQ-KV against ColQwen2-fresh and full context on the three second readers. This appendix adds the remaining arm family at the same $b = 5\%$ budget (Table 6), with page-set semantics identical to the Qwen3-VL-8B panel: the same cached CLIP and ColQwen2 page scores, the same commit-once aggregation over the first $M = \min(3, k)$ queries, the same seed-0 random and truncation controls, and the gold-page oracle read directly. The read path is identical to each port’s ColQwen2 comparator arm, so every number is within-reader comparable. SnapKV arms are computed from each reader’s *own* question-preview attention, aggregated to pages with the panel’s pooling, so SnapKV-fresh and SnapKV-stale are genuinely per-reader rather than proxied from the development reader. All six

Group	Arm	LDU	MMLB
ours (page level)	PAQ (fresh)	0.386	0.258
	attn_max (alt-agg)	0.337	0.152
isolation	same-scorer commit-once	0.128	0.095
	gold-informed static	0.278	0.154
retrieval	ColQwen2-fresh	0.512	0.328
	ColQwen2-stale	0.216	0.138
	CLIP-fresh	0.155	0.164
reference	SnapKV-fresh	0.298	0.157
	full-context	0.546	0.397
	oracle-evidence	0.609	0.431
query-blind	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (proxy)	0.045	0.031
	LOOK-M ($\rightarrow$ H2O)	0.045	0.039
	KVzip (proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

Table 5: Every executed arm, accuracy at $b=5\%$, per cell (LDU = LongDocURL, MMLB = MMLongBench-Doc), on the 903-query confirmatory panel (paq_verdict). Query-aware arms marked fresh re-select per query; stale arms commit once per document. Cohort note: this 903-query confirmatory panel is the page-level study with the *uncorrected* ColQwen2 loader (ColQwen2-fresh MMLB 0.328); the below-page main comparison (Table 1) is a distinct cohort (444 decoded / 440 paired) scored with the *corrected* loader (ColQwen2-fresh MMLB 0.3282, §4.4). The loader correction is immaterial, so the shared reference rows agree to rounding.

reader-cell runs are complete (907 queries by seven arms per reader, zero skipped rows), and everything in this appendix is development-grade.

Quantitatively: SnapKV-fresh, the strongest training-free baseline, sits below PAQ-KV by -0.061 $[-0.096, -0.025]$ (MMLB) and -0.083 $[-0.144, -0.031]$ (LDU) on Qwen3-VL-4B, by -0.143 on both Qwen2.5-VL-7B cells, and by -0.099 and -0.142 on InternVL3.5-8B. Its absolute level falls from 0.303 to 0.127 to 0.089 across the three readers on MMLongBench-Doc, and from 0.413 to 0.252 to 0.077 on LongDocURL, the same ordering as the recovery gradient of §5.5. Committing either scorer once collapses it on every reader (Table 6). On InternVL3.5-8B the gold-page oracle exceeds full context by $+0.052$ $[+0.013, +0.094]$ on MMLB and $+0.045$ $[+0.007, +0.088]$ on LDU, so evidence conversion is not that reader’s bottleneck; finding the evidence is.

C MMDocIR: Below-Page Selection Quality

MMDocIR (Dong et al. 2025a) provides expert gold at *layout* granularity: paragraphs, tables, figures, and charts as boxes on the page. This is ground truth at exactly PAQ-KV’s below-page granularity, which neither QA benchmark offers. On a fresh-documents-only pin (80 documents, 538 queries, 641 gold layouts; the same exclusion audit as Appendix F), we score PAQ-KV’s selected 64-token blocks ge-

ometrically against the gold boxes: block token spans are mapped through the reader’s own patch grid to page-pixel rectangles, and we measure gold-layout area coverage, layout hit rate, and gold-page recall at $b \in \{5, 10\}\%$ against random-block, truncation-block, and page-granularity comparators at matched token budget (Table 7). No decoding is involved, so the analysis isolates the selector from QA conversion entirely.

Three observations. The selector finds sub-page evidence: at a realized 5.3% of context, PAQ-KV’s blocks cover 0.689 of gold-layout area, where random blocks at the same budget cover 0.056. This is the study’s first conversion-free measurement of the selection signal against human-annotated sub-page ground truth. Below-page granularity buys reach rather than raw area: the page-granularity arm, using identical attention scores aggregated to pages, covers the same gold area but needs a realized 7.7% of context to do it under whole-page rounding, and at that higher spend still touches fewer distinct gold layouts ($+0.140$ $[+0.097, +0.170]$ in PAQ-KV’s favor) and fewer gold pages. Blocks spread the same budget across more of the question’s evidence, which is the behavior the cross-page QA advantage of Appendix H suggested, now measured geometrically. Finally, selecting every gold page in full would cost a realized 5.6% of context on this pin, essentially PAQ-KV’s own spend, so a perfect page-level selector could reach full coverage at matched budget; PAQ-KV recovers 69% of that area with no supervision, from the reader’s own attention, and the remaining gap is the selection headroom that the oracle contrasts of Appendix B price in accuracy terms.

D The Page-Level System in Detail

The name PAQ abbreviates “Prefill once, Ask many Queries”; unsuffixed PAQ denotes the earlier page-level router (v1) detailed in this appendix, PAQ-T its expert-head variant, and PAQ-KV the below-page KV-mask method of the main text. All numbers in this section are on the 903-query confirmatory cohort (page-level arms), where ColQwen2-fresh scores 0.512/0.328 (LDU/MMLB) with the uncorrected loader; it is distinct from the below-page cohort used in the main comparison (where the corrected-loader ColQwen2-fresh MMLB is 0.3282, §4.4). The loader correction is immaterial (§4.4), so the page-level parity conclusions are unaffected.

The page router (PAQ v1). Pages are rendered at coarse resolution ($\approx 3.5K$ visual tokens total, i.e. ≈ 70 tokens/page on a 50-page document) and prefilled once; per query, the question rows’ attention (mean over heads, layers, and question tokens) scores pages; the top pages at budget b are re-rendered at full resolution and read. This is page routing rather than KV compression, and it claims no memory savings.

Comparison against trained retrieval. At $b = 5\%$, ColQwen2-fresh scores 0.512/0.328 vs. PAQ 0.386/0.258: pooled -0.099 $[-0.129, -0.072]$, a loss of roughly 10 points that holds on every stratum we examined (Figure 4). Gold-page recall is consistent with a coverage contribution:

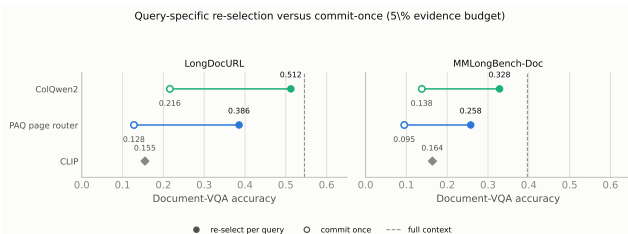


Figure 4: Two observations from the page-level system, per cell: (i) ColQwen2-fresh outperforms the page-level attention router (a loss of roughly 10 points for the router); (ii) *both* scorers collapse when committed once, so reversibility is a general property of the multi-query regime rather than being scorer-specific. PAQ here is the earlier page-level router of this appendix, not PAQ-KV; the below-page PAQ-KV comparison at matched budget is in Table 1.

ColQwen2-fresh 0.664/0.536 vs. PAQ 0.494/0.443. Both scorers collapse when committed once (ColQwen2 stale 0.216/0.138), reproducing the reversibility pattern with a second scorer family.

PAQ-T: expert-head recall and the parity accuracy evaluation. Scoring with a single expert (layer, head) chosen by held-out-document cross-validation (CAPS’s technique) lifts held-out gold-page recall@5% to 0.683 ($n=451$) / 0.626 ($n=425$), meeting or exceeding ColQwen2’s in-harness 0.664/0.536; mean aggregation at higher scoring resolution moves almost nothing (0.494/0.443 $\rightarrow$ 0.500/0.452), so the head, not the resolution, is the lever. All selected fold heads sit in layer 21. The pre-registered accuracy evaluation (bars frozen before the run; same reader, render policy, and scorer as the cached baseline rows; only the selected page set differs) lands at parity: PAQ-T 0.518/0.357 vs. ColQwen2-fresh 0.512/0.328; deltas $+0.006$ [$-0.018, +0.030$] and $+0.028$ [$+0.001, +0.055$]; pooled $+0.017$ [$-0.000, +0.035$]. This is non-inferior under the frozen lower-bound margin (> -0.03) on both cells but not the pre-registered improvement criterion (all lower bounds > 0), and we do not present the marginal per-cell interval as evidence of superiority. The initially better-looking complete-case verdict (pooled $+0.023$, lower bound $+0.005$) was blocked in review for informative missingness: the $\approx 4\%$ of queries whose high-resolution scoring ran out of memory had cached baseline accuracy 0.62, far above the cell means; a conservative reduced-resolution recovery of 31/35 missing rows (degrading only the PAQ-T arm, never the baseline) erased the pooled edge. Two caveats carry over: head selection is cross-validated *benchmark-supervised* model selection (not zero-shot; the externally selected L21.H18 of the negative transfer-gate ablation later removed this caveat from the head’s existence, though not from its benchmark-tuned use), and the expert-head technique is CAPS’s, so the residual novelty at page level is only the amortized one-prefill-many-queries architecture.

E Additional Reversibility Detail

Figure 5 summarizes the confirmatory below-page gate and its supporting replications. For the confirmatory gate: *missingness* — all 907 pinned queries completed (zero skips), so the pre-registered adversarial imputation guard (fresh $\rightarrow 0$, comparators $\rightarrow 1$ on any incomplete query) never binds; *budgets* — all three arms realize a matched $\approx 6.0\%$ of context. Supporting analyses from the historical page-level run (pre-fix cohort, labelled): (i) *answerable subset* (secondary, since it conditions on a model outcome): on queries where oracle-evidence scores exactly 1 ($n = 407$), fresh 0.583 vs. commit-once 0.182 ($+0.395$ [$+0.324, +0.460$]) and vs. the static 0.382 ($+0.204$ [$+0.135, +0.266$]); (ii) *budget curves*: the fresh page-level selector dominates both static comparators at every budget in $\{2.5, 5, 10\}\%$ on both cells.

Cache-pollution quantification. The in-place cache bug of the main text was quantified before being fixed: on a 6-documents-per-cell diagnostic (87 query-rows), the bug inflates gold-page recall by $\approx +0.05$ for a fixed scoring head and shifts selections (mean top- B Jaccard clean vs. polluted 0.82, drifting from 1.00 at q_0 to 0.61 at q_6), as expected for cumulative contamination. The recall gate behind the PAQ-T numbers was regenerated clean before any accuracy read (clean held-out expert recall 0.683/0.626 vs. polluted 0.674/0.628), and the clean cross-validation picks a more robust head. Absolute fresh-arm accuracy in the pre-fix confirmatory run carries ≈ 0.04 inflation (clean — polluted -0.041 [$-0.077, -0.003$] on the 24-document check); the isolation contrasts cancel the bug and stand.

F Reproducibility and Protocol Notes

Pre-registration discipline. Every gate in this study was frozen (bars, comparators, budgets, imputation policy) before its run: the two-comparator isolation gate; the PAQ-T accuracy-evaluation bars (non-inferiority/improvement/inferior margins, per-cell power floors, adversarial missingness imputation); the per-head-mixture transfer gate; and the PAQ-KV decode-identity kill criterion, screen gates, per-cell improvement bars, and sharp-scorer (V1) gate. Two early positives did not survive adversarial review and were retracted before this draft: a first isolation gate against the eviction family (invalidated by a weak-static artifact and replaced by the two-comparator gate), and an early “outperforms the gold-page oracle” claim (corrected to “matches” at power). The commitment-pressure mechanism hypothesis failed its formal test and is reported as exploratory.

Instrumentation. All significance statements use one instrument: the document-cluster paired bootstrap (10,000 draws, seed 0, percentile intervals; equal-cell pooling), unit-tested alongside the gate and pairing logic. Evaluation pins are sha-256 checksummed and byte-verified before use; the screen pin is a strict prefix of the confirmatory pin, and independent-extension analyses use the 50 new documents per cell only. Figures are generated from the verdict artifacts by a single script; nothing is hand-entered. Per-query, per-arm rows, frozen-bar metadata, verdict files, and the adver-

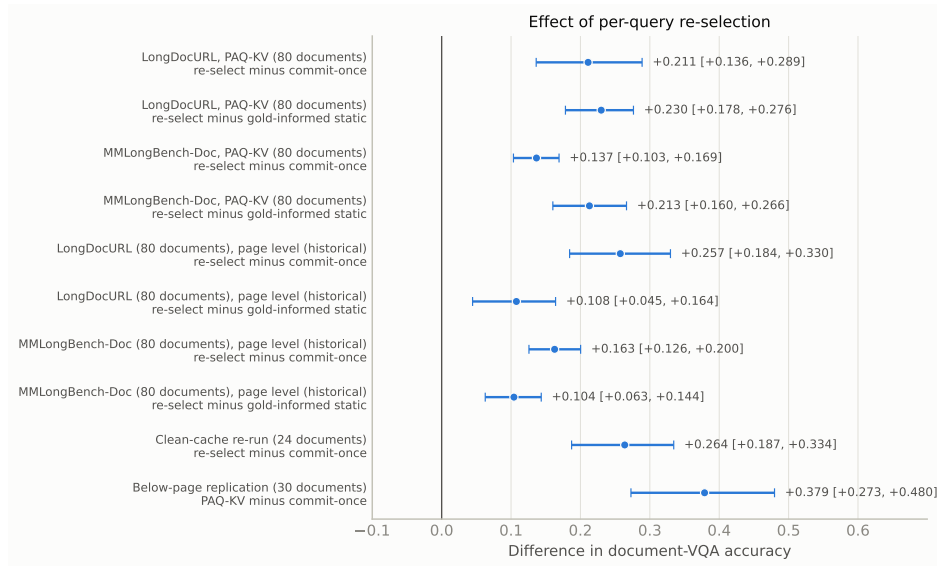


Figure 5: Document-clustered two-sided 95% intervals for every executed isolation contrast. The leading four rows are the confirmatory below-page gate (PAQ-KV as the fresh arm, full 80-document pins, post-fix code); below them, the historical page-level gate, the clean-cache re-run, and the 30-document below-page replication. Positive differences favor per-query re-selection; every interval excludes zero.

serial review trail are retained and will be released with the code. The descriptive re-slices in Appendix H, the realized-budget figures, the amortized-latency column of Table 4, and the budget sweep of Appendix I are additional reuses of the development pin and carry no pre-frozen bars.

Sealed-test protocol. The 80-document confirmatory pin has been reused across this study and is therefore development-only. The fresh sealed set (zero overlap with the pin; 45 MMLongBench-Doc documents / 244 queries, plus 73 LongDocURL documents left unevaluated as out of scope) was evaluated *exactly once* under a guarded, frozen harness: the confirmatory bar (complete-case document-clustered CI lower bound > 0 , with an adversarial worst-case-imputation fallback binding only under any missingness), the corrected-loader comparator, the 5% budget, the seed, and sha-256 hashes of all eight code components plus the sealed pin were frozen before the run; the harness refuses the sealed pin absent an explicit authorization flag and a typed confirmation, aborts on any component-hash mismatch, and runs a blind non-scoring preflight that can abort without consuming the test. The single run produced the strongest claim in this paper: PAQ-KV@5% 0.396 vs. ColQwen2-fresh 0.320, +0.076 with document-clustered 95% CI [+0.029, +0.122] (244/244 complete, zero skips, robust to adversarial imputation; §5.2). The verdict, per-query rows, and a provenance manifest (pin and component hashes, resolved model revisions) are retained and released with the code.

Extended-benchmark overlap audit. The two added corpora are corpus-adjacent to ours: MMDocIR was sourced from existing DocVQA collections including MMLongBench-Doc, and MMDocRAG was built over the

MMDocIR document pool. A document-name audit against our pins found substantial overlap, including, name for name, the entire sealed MMLongBench-Doc cell inside MMDocIR’s pool. Every document overlapping either the development or the sealed pins was excluded from the new pins before any experiment ran, so the sealed reserve remains untouched at the document level; only document names were read from the sealed pin file, and the audit artifact is retained. The MMDocRAG pin’s protocol deviations are disclosed here: 62 qualifying documents is below the study’s 80-document floor; queries are capped at 12 per document by seed-0 sampling (585 of 1,160 kept), a cap the main pins never needed; pages render at 144 DPI, matching the existing pins’ pixel scale; gold pages are the union of page identifiers over each query’s gold quotes, with the 0-based convention verified by text-fragment matching on 32 of 32 probes; and answers are scored by the study’s shared document-VQA evaluator. The extended cells reuse the study’s frozen primitives (prefill, preview, selection, decode, the corrected ColQwen2 loader, the shared scorer, and the document-cluster paired bootstrap) but froze no bars, ran each cell once, and constructed the pins on the day of the runs.

G Additional Method and Gate Tables

H Selection Anatomy on MMLongBench-Doc

These are descriptive re-slices of the retained per-query development-pin rows (444 decoded / 440 paired, corrected ColQwen2 loader throughout); no pre-frozen bar attaches to them.

Algorithm 1 PAQ-KV: per-query reversible below-page KV-block selection with a 4D restricted-attention mask

Require: document pages rendered as in the full-context arm ($\leq 64K$ tokens); queries $q_1, \dots, q_k$; nominal budget $b = 5\%$; frozen VLM

- 1: prefill the document once; retain the full KV cache C (context length n_{ctx}); record page-aligned 64-token block spans $B_1, \dots, B_m$ and the always-keep scaffold set $\mathcal{A}$
- 2: **for** each query q_j **do**
- 3: forward the tokenized question over C ; capture question-to-context attention via per-layer hooks; **crop** C back to n_{ctx}
- 4: $\text{score}(B_i) \leftarrow$ aggregated question-row attention mass on block B_i 's positions
- 5: $S_j \leftarrow$ top blocks by score at the nominal budget $b \cdot n_{\text{ctx}}$
- 6: build the 4D additive mask M_j : decode rows may attend to $\mathcal{A} \cup S_j \cup q_j$ generated tokens; all other context positions get $-\infty$
- 7: decode the answer from C under M_j ; no gathering, so K/V positions and M-RoPE indices are unchanged
- 8: discard M_j ; C is intact for q_{j+1}
- 9: **end for**

Gold-page recall vs. accuracy. At the nominal 5% budget, two recall notions bracket what each method hands the reader. Coverage-weighted (the fraction of gold-page *tokens* the reader may attend; the budget-comparable notion, since ColQwen2 keeps whole pages): ColQwen2 0.539 vs. PAQ-KV 0.344, a deficit of -0.195 $[-0.236, -0.152]$ for PAQ-KV. Any-block-touch (a gold page counts as recalled if any selected 64-token block overlaps it): PAQ-KV 0.912, but this notion flatters block granularity, since PAQ-KV's selections scatter over a median of 10 pages per query. So the retriever delivers substantially more gold-evidence tokens at matched budget, touches fewer of the right pages, and still loses the accuracy comparison by $+0.077$ (Table 1): recall-optimized selection is not accuracy-optimized selection, consistent with the sharp-scorer negative result (§5.6).

Evidence-type stratification. Table 10 stratifies the main paired comparison by the benchmark's evidence annotations. The cross-page stratum roughly doubles the single-page contrast ($+0.109$ vs. $+0.054$), and the source strata whose intervals exclude zero (chart, table, layout) are those whose evidence a page-level retriever must fetch whole even when only a small region answers the question.

I Full-Pin Budget Sweep

The improvement is not a 5%-only artifact, and on the full pin the earlier screen-only caveats resolve favorably. PAQ-KV's full-pin budget curve is monotone (0.389/0.404/0.405 at 2.5/5/10%), so the mild non-monotonicity previously observed on the 157-query screen subset (0.397/0.404/0.398) was small-sample noise. Against a ColQwen2 comparator re-read at the *matched* page budget (Table 11), PAQ-KV is clearly ahead at 2.5%, ahead by the pre-registered margin at 5%, and ahead on point estimate at 10% with an interval that

just includes zero: the margin narrows as budget grows, because the retriever gains more from additional whole pages than PAQ-KV gains from additional blocks. Against the fixed ColQwen2@5% comparator, every PAQ-KV budget remains ahead ($+0.064$, $+0.077$, $+0.078$ at 2.5/5/10%, all lower bounds positive; 440 paired queries).

J Additional Limitations Detail

LongDocURL conversion-bound decomposition. A re-analysis of the executed rows suggests the shortfall there lies largely past selection: of the 463 rows, $\approx 39\%$ fail under both PAQ-KV and the gold-page oracle (zero even when handed the native-resolution gold pages), only $\approx 10\%$ are oracle-right/PAQ-KV-wrong (the selection-capturable pool), and 2.2% are PAQ-KV-right/oracle-wrong, so the oracle is not a strict ceiling; that no selection method can rescue these queries is an observation about this dataset, not a general rule. Native resolution already saturates the prefill on 79/80 documents, and large documents are a flat, low-leverage half. Among pure-selection levers, only a marginal oracle-chooser union passes (lower bound $+0.003$; not deployable); only a PAQ-KV $\cup$ ColQwen2 hybrid passes comfortably (lower bound $+0.061$), and we exclude it as not a standalone method. We therefore offer the dispersed-vs-conversion-bound split as a hypothesis from two datasets.

Base-model ceiling counts. A large fraction of queries receive no credit even given exactly the annotated gold pages: the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of confirmatory queries and a full 1 on only 45.1%. Pairing controls for shared query difficulty; the oracle is not a strict ceiling (a selected page set can score above the annotated gold set on individual queries).

Remaining protocol caveats. The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order (order-permutation and M -sensitivity checks remain to be run; the order-free static partially covers this), and the page-level confirmatory run predates the cache fix (§4.4); its isolation *contrasts* were verified robust, but its absolute fresh-arm levels carry ≈ 0.04 inflation on the 24-document check.

Arm	LDU	MMLB
Qwen3-VL-4B (same architecture)		
PAQ-KV@5% (ours; cached)	0.4966	0.3640
ColQwen2-fresh (cached)	0.4908	0.3336
full-context (ref.; cached)	0.5024	0.3480
SnapKV-fresh	0.4133	0.3034
SnapKV-stale	0.1332	0.0873
CLIP-fresh	0.1545	0.1805
ColQwen2-stale	0.2105	0.1576
random	0.0684	0.0296
truncation	0.0322	0.0423
gold-page oracle (ref.)	0.5839	0.4373
Qwen2.5-VL-7B (same family, prior gen.)		
PAQ-KV@5% (ours; cached)	0.3953	0.2698
ColQwen2-fresh (cached)	0.4728	0.2623
full-context (ref.; cached)	0.4558	0.3237
SnapKV-fresh	0.2519	0.1265
SnapKV-stale	0.0574	0.0342
CLIP-fresh	0.1591	0.1354
ColQwen2-stale	0.2047	0.1245
random	0.0609	0.0253
truncation	0.0343	0.0394
gold-page oracle (ref.)	0.5380	0.3219
InternVL3.5-8B (cross-family, Qwen3 LM)		
PAQ-KV@5% (ours; cached)	0.2186	0.1876
ColQwen2-fresh (cached)	0.3081	0.2542
full-context (ref.; cached)	0.2961	0.2637
SnapKV-fresh	0.0770	0.0889
SnapKV-stale	0.0537	0.0492
CLIP-fresh	0.1121	0.1498
ColQwen2-stale	0.1549	0.1068
random	0.0675	0.0377
truncation	0.0293	0.0385
gold-page oracle (ref.)	0.3408	0.3159

Table 6: Baseline arm family on the three second readers at the 80-document development pins (LDU = LongDocURL, MMLB = MMLongBench-Doc), $b = 5\%$. The PAQ-KV, ColQwen2-fresh, and full-context rows are the cached values from the cross-reader ports (identical to Table 3); the remaining arms are new runs with matched page-set semantics. Development-grade.

Arm	layout cov.	layout hit	page recall	realized b
nominal budget $b = 0.05$				
PAQ-KV blocks (ours)	0.689	0.834	0.903	5.3%
page level (same scorer)	0.693	0.694	0.694	7.7%
random blocks	0.056	0.141	0.449	5.3%
truncation blocks	0.079	0.087	0.095	5.4%
nominal budget $b = 0.1$				
PAQ-KV blocks (ours)	0.778	0.895	0.960	10.3%
page level (same scorer)	0.791	0.792	0.792	12.2%
random blocks	0.114	0.273	0.720	10.2%
truncation blocks	0.125	0.127	0.138	10.3%
whole gold pages (upper ref.)	1.000	1.000	1.000	5.6%

Table 7: Selection quality against MMDocIR’s sub-page gold layouts on the 80-document fresh pin (538 queries). “Coverage” is the fraction of gold-layout area covered by selected blocks on gold pages; “hit” the fraction of gold layouts touched at all; “gold-page recall” the fraction of gold pages receiving any block. Development-grade.

	LongDocURL	MMLongBench
Documents	80	80
Complete queries at gate	463	444
Gold-bearing queries / doc	5.79	5.50
Median (mean) union pages	44 (45.0)	28 (36.5)

Table 8: The confirmatory evaluation pin. “Complete queries” are rows where every gated arm scored; the below-page gate completed all 907 pinned queries with zero skips, so no imputation was needed (the historical page-level run had 4 timeout-missing queries, immaterial under worst-case imputation).

Contrast at $b=5\%$	LongDocURL	MMLongBench-Doc	pooled (equal-cell, secondary)
fresh — same-scorer commit-once	+0.211 [+0.136, +0.289]	+0.137 [+0.103, +0.169]	+0.174 [+0.132, +0.215]
fresh — gold-informed static	+0.230 [+0.178, +0.276]	+0.213 [+0.160, +0.266]	+0.221 [+0.184, +0.256]
arm means (fresh / commit-once / static)	0.531 / 0.320 / 0.301	0.405 / 0.268 / 0.192	—
Verdict	REVERSIBILITY ISOLATED (all $LB_{95} > 0$, both cells)		

Table 9: Confirmatory below-page isolation at $b=5\%$ on all 907 pinned queries (accuracy; PAQ-KV as the fresh arm; post-fix code; document-cluster paired bootstrap; contrast rows report Δ with 95% CIs). The two-comparator bar was frozen and committed before the run; the per-cell contrasts are the pre-registered primary, the pooled column is secondary.

Stratum	n docs PAQ-KV ColQ.				Δ (95% CI)
Chart	102	35	0.409	0.311	+0.098 [+0.025, +0.176]
Figure	156	53	0.393	0.333	+0.060 [−0.020, +0.143]
Layout	47	32	0.490	0.338	+0.152 [+0.032, +0.252]
Pure text	144	52	0.356	0.308	+0.048 [−0.010, +0.104]
Table	120	38	0.353	0.258	+0.096 [+0.033, +0.163]
Single-page	256	74	0.526	0.471	+0.054 [+0.001, +0.108]
Cross-page	184	70	0.238	0.129	+0.109 [+0.052, +0.170]

Table 10: Paired PAQ-KV@5% — ColQwen2-fresh (corrected loader) by MMLongBench-Doc evidence stratum on the 440-query paired cohort (document-cluster paired bootstrap). Evidence-source labels are the benchmark’s own and are multi-label (a query counts under each listed source); single- vs. cross-page splits on the number of annotated gold pages. The improvement concentrates on cross-page evidence and on chart/table/layout sources; these are descriptive development-pin strata, not pre-registered claims.

Budget	PAQ-KV	ColQwen2@ b	Δ (95% CI)
2.5%	0.389	0.291	+0.098 [+0.058, +0.137]
5%	0.404	0.328	+0.077 [+0.040, +0.115]
10%	0.405	0.370	+0.036 [−0.001, +0.072]

Table 11: Full-pin budget sweep on MMLongBench-Doc (80 documents; matched budgets, corrected ColQwen2 loader). The 5% row is the pre-registered main comparison (440 paired queries, Table 1); the 2.5/10% rows are descriptive (444 queries each; the pre-frozen bar was frozen at 5% only). Realized fractions: PAQ-KV 3.5/6.0/10.9% of context, ColQwen2 3.6/4.6/8.6% of pages.